

TE Curve Verification Pipeline Familywise ONNX Report - wave3_2_harmonic_residual_offset

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_harmonic_residual_offset/2026-07-11-11-07-57__track2_wave3_2_harmonic_residual_offset_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_harmonic_residual_offset/2026-07-11-11-07-57__track2_wave3_2_harmonic_residual_offset_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_harmonic_residual_offset/2026-07-11-11-07-57__track2_wave3_2_harmonic_residual_offset_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_harmonic_residual_offset/2026-07-11-11-07-57__track2_wave3_2_harmonic_residual_offset_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave3_2_harmonic_residual_offset` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave3_2_harmonic_residual_offset_fw_simplified_setpoints</code>	<code>2026-07-10-23-36-51_te_wave3_2_harmonic_residual_offset_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave3_2_harmonic_residual_offset_bw_simplified_setpoints</code>	<code>2026-07-10-23-52-02_te_wave3_2_harmonic_residual_offset_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave3_2_harmonic_residual_offset_global_simplified_setpoints</code>	<code>2026-07-10-23-18-29_te_wave3_2_harmonic_residual_offset_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/forward/2026-07-10-23-36-51_te_wave3_2_harmonic_residual_offset_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/forward/2026-07-10-23-36-51_te_wave3_2_harmonic_residual_offset_fw_simplified_setpoints/python/harmonic_residual_offset_probe-epoch=121-val_mae=0.00362257.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/backward/2026-07-10-23-52-02_te_wave3_2_harmonic_residual_offset_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/backward/2026-07-10-23-52-02_te_wave3_2_harmonic_residual_offset_bw_simplified_setpoints/python/harmonic_residual_offset_probe-epoch=091-val_mae=0.00361216.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/global/2026-07-10-23-18-29_te_wave3_2_harmonic_residual_offset_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/global/2026-07-10-23-18-29_te_wave3_2_harmonic_residual_offset_global_simplified_setpoints/python/harmonic_residual_offset_probe-epoch=138-val_mae=0.00362371.ckpt</code>

Aggregate Metrics

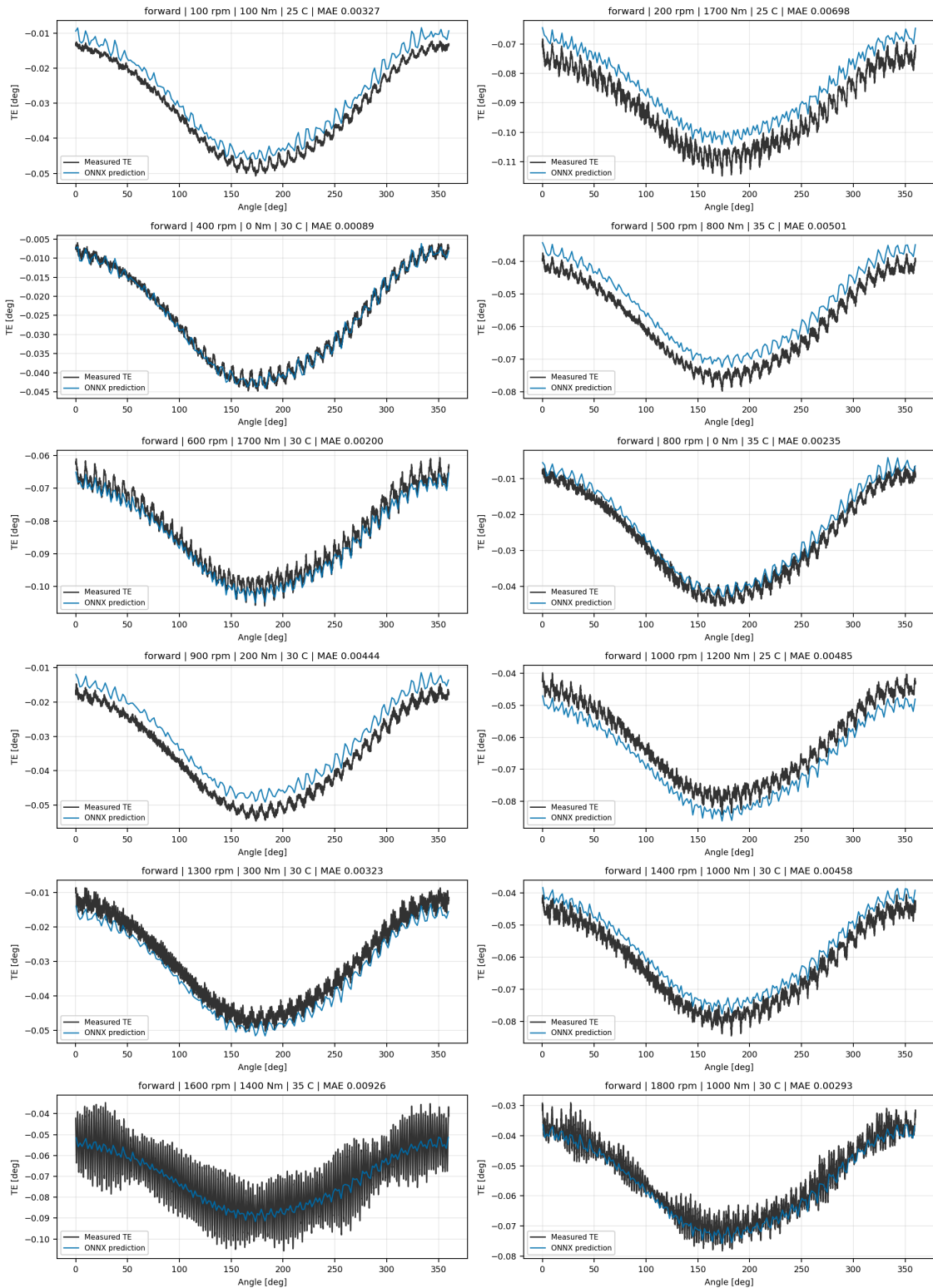
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003257	0.003530	7.613	12.571
backward	97	0.003554	0.003876	8.258	13.391
global	194	0.003405	0.003721	7.935	13.730

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000323	0.002861	0.001256	0.003393
backward	0.000063	0.002954	0.001557	0.003343
global	-0.000206	0.002883	0.001466	0.003029

Forward 12-Curve Page

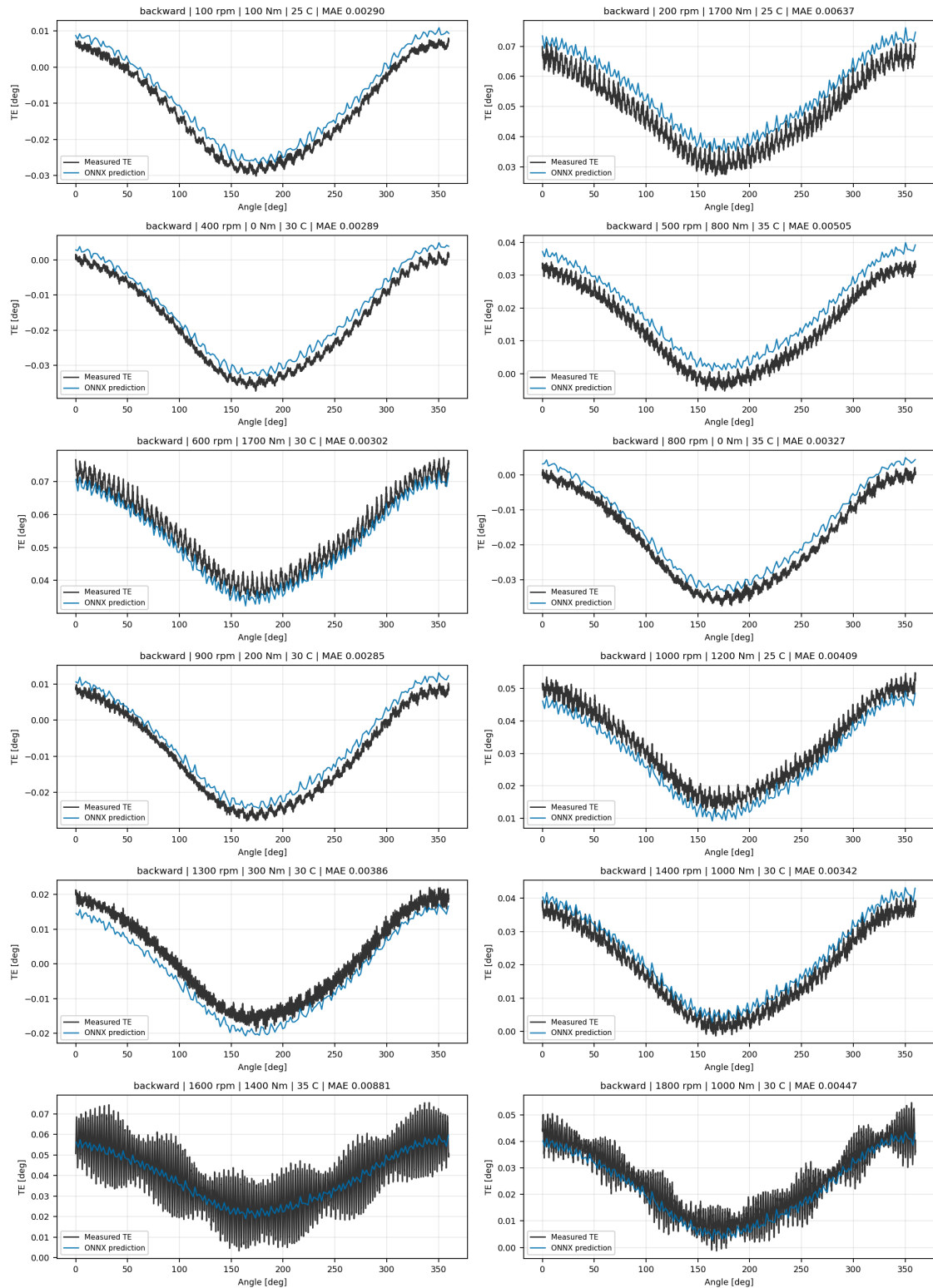
Simplified Dataset + Setpoints | wave3_2_harmonic_residual_offset | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

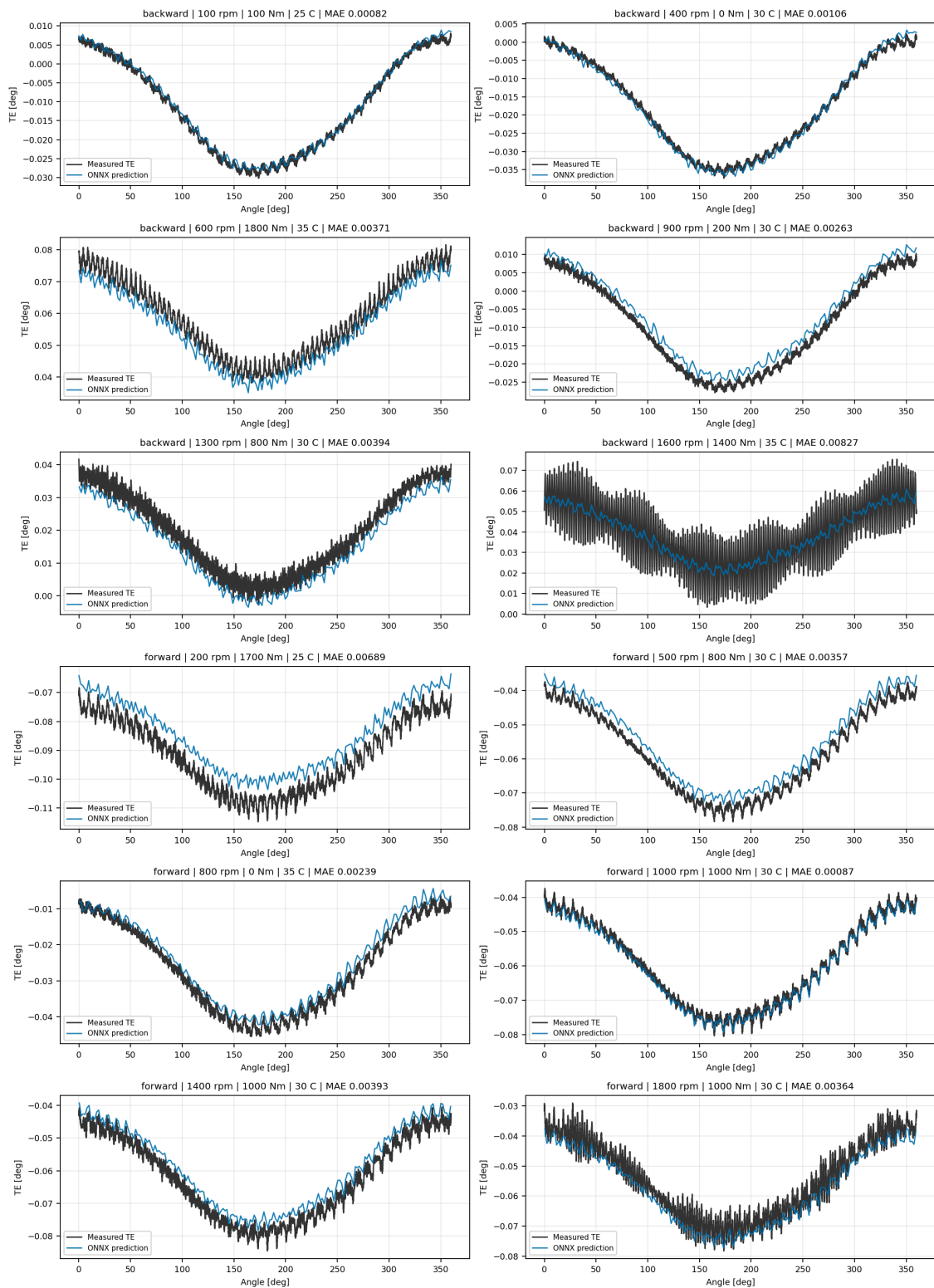
Simplified Dataset + Setpoints | wave3_2_harmonic_residual_offset | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave3_2_harmonic_residual_offset | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave3_2_harmonic_residual_offset ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_2_harmonic_residual_offset_fw_polished_setpoints	2026-07-11-00-52-59_te_wave3_2_harmonic_residual_offset_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave3_2_harmonic_residual_offset_bw_polished_setpoints	2026-07-11-01-14-42_te_wave3_2_harmonic_residual_offset_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave3_2_harmonic_residual_offset_global_polished_setpoints	2026-07-11-00-24-33_te_wave3_2_harmonic_residual_offset_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/forward/2026-07-11-00-52-59_te_wave3_2_harmonic_residual_offset_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/forward/2026-07-11-00-52-59_te_wave3_2_harmonic_residual_offset_fw_polished_setpoints/python/harmonic_residual_offset_probe-epoch=085-val_mae=0.00188607.ckpt
backward	models/polished_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/backward/2026-07-11-01-14-42_te_wave3_2_harmonic_residual_offset_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/backward/2026-07-11-01-14-42_te_wave3_2_harmonic_residual_offset_bw_polished_setpoints/python/harmonic_residual_offset_probe-epoch=120-val_mae=0.00193287.ckpt
global	models/polished_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/global/2026-07-11-00-24-33_te_wave3_2_harmonic_residual_offset_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_2_harmonic_residual_offset/global/2026-07-11-00-24-33_te_wave3_2_harmonic_residual_offset_global_polished_setpoints/python/harmonic_residual_offset_probe-epoch=124-val_mae=0.00190522.ckpt

Aggregate Metrics

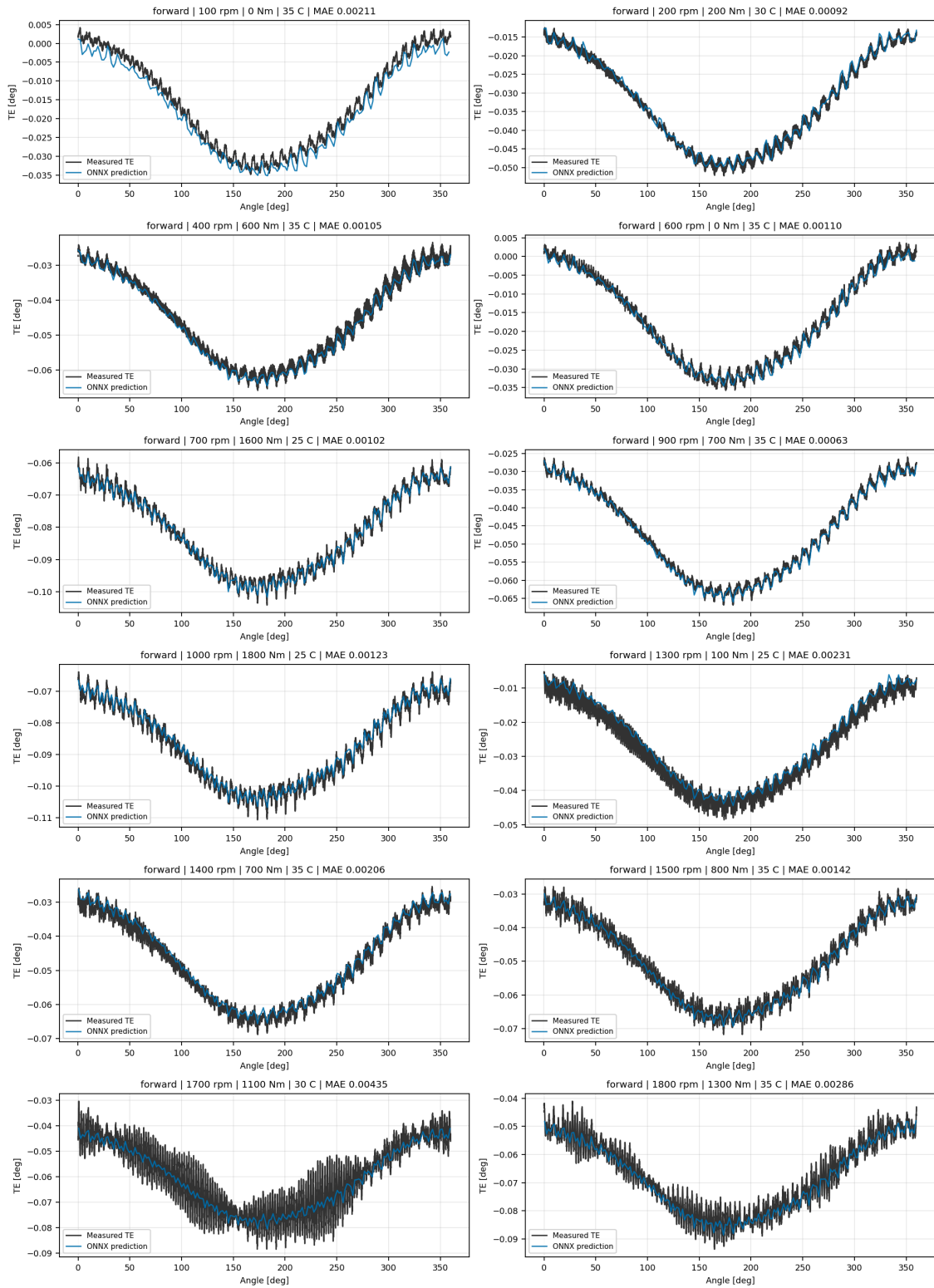
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001937	0.002305	4.262	10.396
backward	94	0.002514	0.002960	4.594	10.864
global	194	0.002220	0.002623	4.417	10.781

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000001	0.000950	0.001582	0.003934
backward	0.000378	0.000977	0.002127	0.005713
global	0.000191	0.001008	0.001818	0.004566

Forward 12-Curve Page

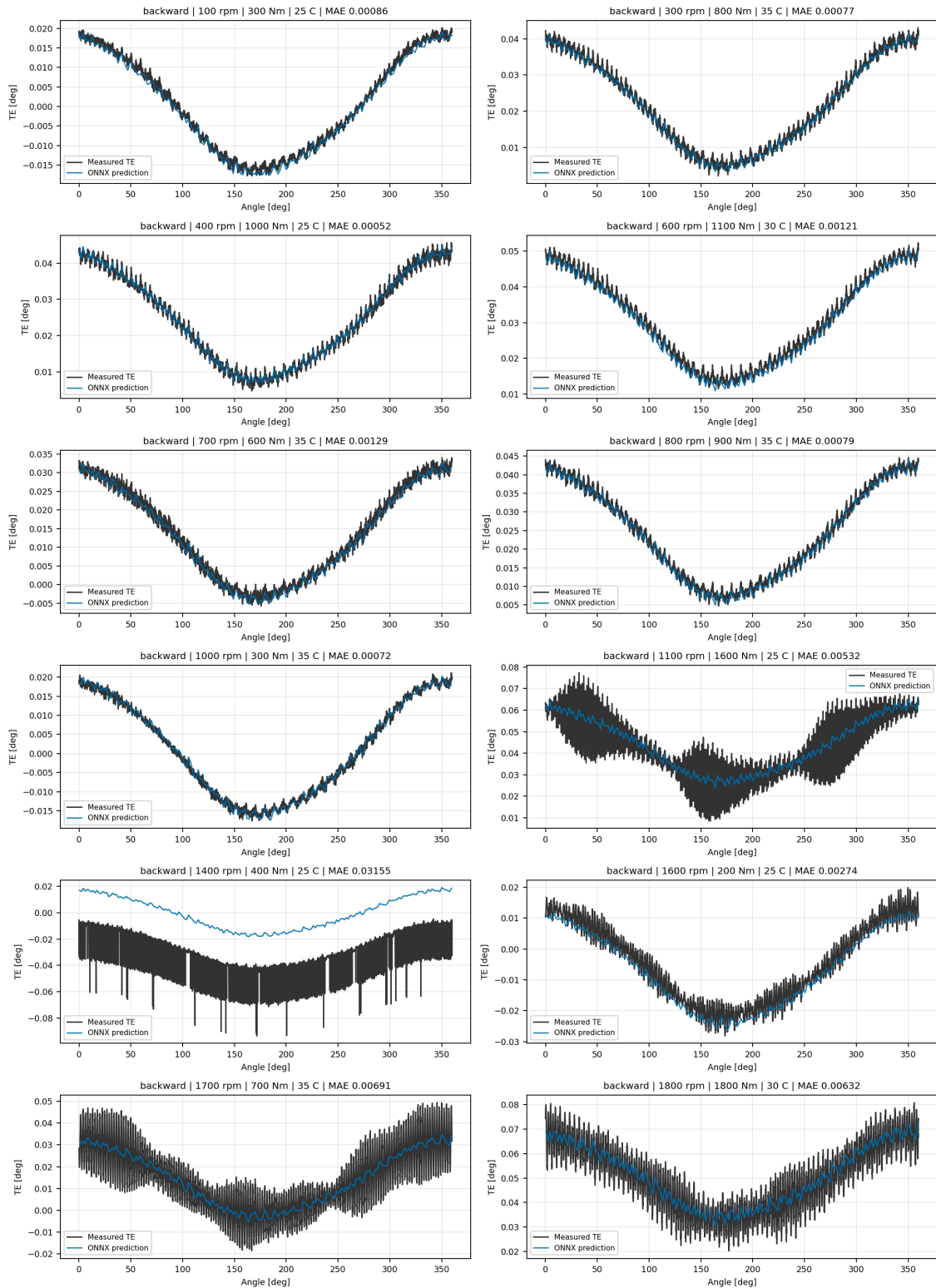
Polished Dataset + Setpoints | wave3_2_harmonic_residual_offset | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

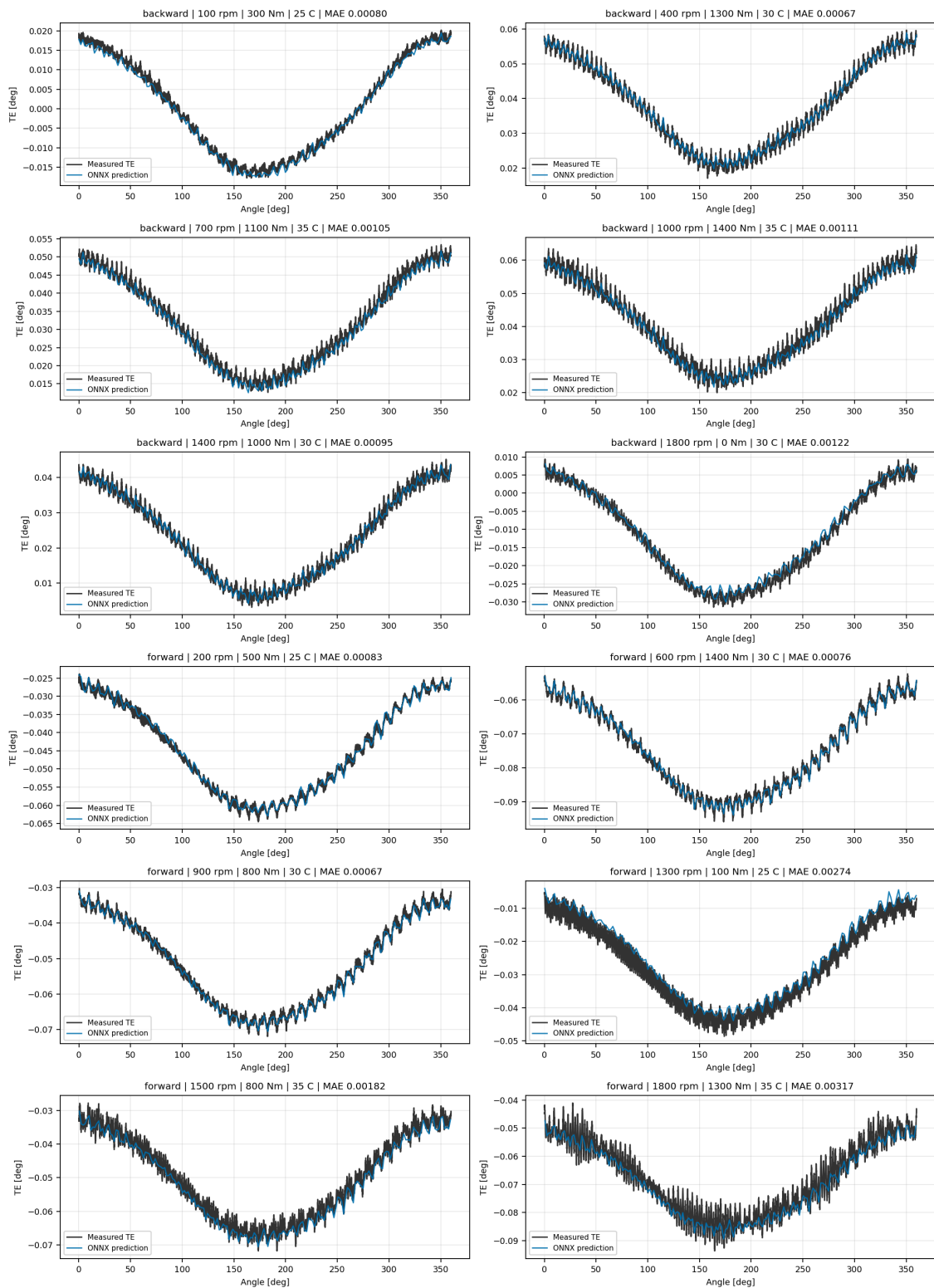
Polished Dataset + Setpoints | wave3_2_harmonic_residual_offset | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave3_2_harmonic_residual_offset | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave3_2_harmonic_residual_offset ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_2_harmonic_residual_offset_fw_polished_actual_values	2026-07-11-05-29-51_te_wave3_2_harmonic_residual_offset_fw_polished_actual_values	polished_point_v1
backward	te_wave3_2_harmonic_residual_offset_bw_polished_actual_values	2026-07-11-06-13-40_te_wave3_2_harmonic_residual_offset_bw_polished_actual_values	polished_point_v1
global	te_wave3_2_harmonic_residual_offset_global_polished_actual_values	2026-07-11-04-51-02_te_wave3_2_harmonic_residual_offset_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave3_2_harmonic_residual_offset/forward/2026-07-11-05-29-51_te_wave3_2_harmonic_residual_offset_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_2_harmonic_residual_offset/forward/2026-07-11-05-29-51_te_wave3_2_harmonic_residual_offset_fw_polished_actual_values/python/harmonic_residual_offset_probe-epoch=205-val_mae=0.00184965.ckpt
backward	models/polished_dataset/actual_values/exported/wave3_2_harmonic_residual_offset/backward/2026-07-11-06-13-40_te_wave3_2_harmonic_residual_offset_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_2_harmonic_residual_offset/backward/2026-07-11-06-13-40_te_wave3_2_harmonic_residual_offset_bw_polished_actual_values/python/harmonic_residual_offset_probe-epoch=205-val_mae=0.00185299.ckpt
global	models/polished_dataset/actual_values/exported/wave3_2_harmonic_residual_offset/global/2026-07-11-04-51-02_te_wave3_2_harmonic_residual_offset_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_2_harmonic_residual_offset/global/2026-07-11-04-51-02_te_wave3_2_harmonic_residual_offset_global_polished_actual_values/python/harmonic_residual_offset_probe-epoch=201-val_mae=0.00183635.ckpt

Aggregate Metrics

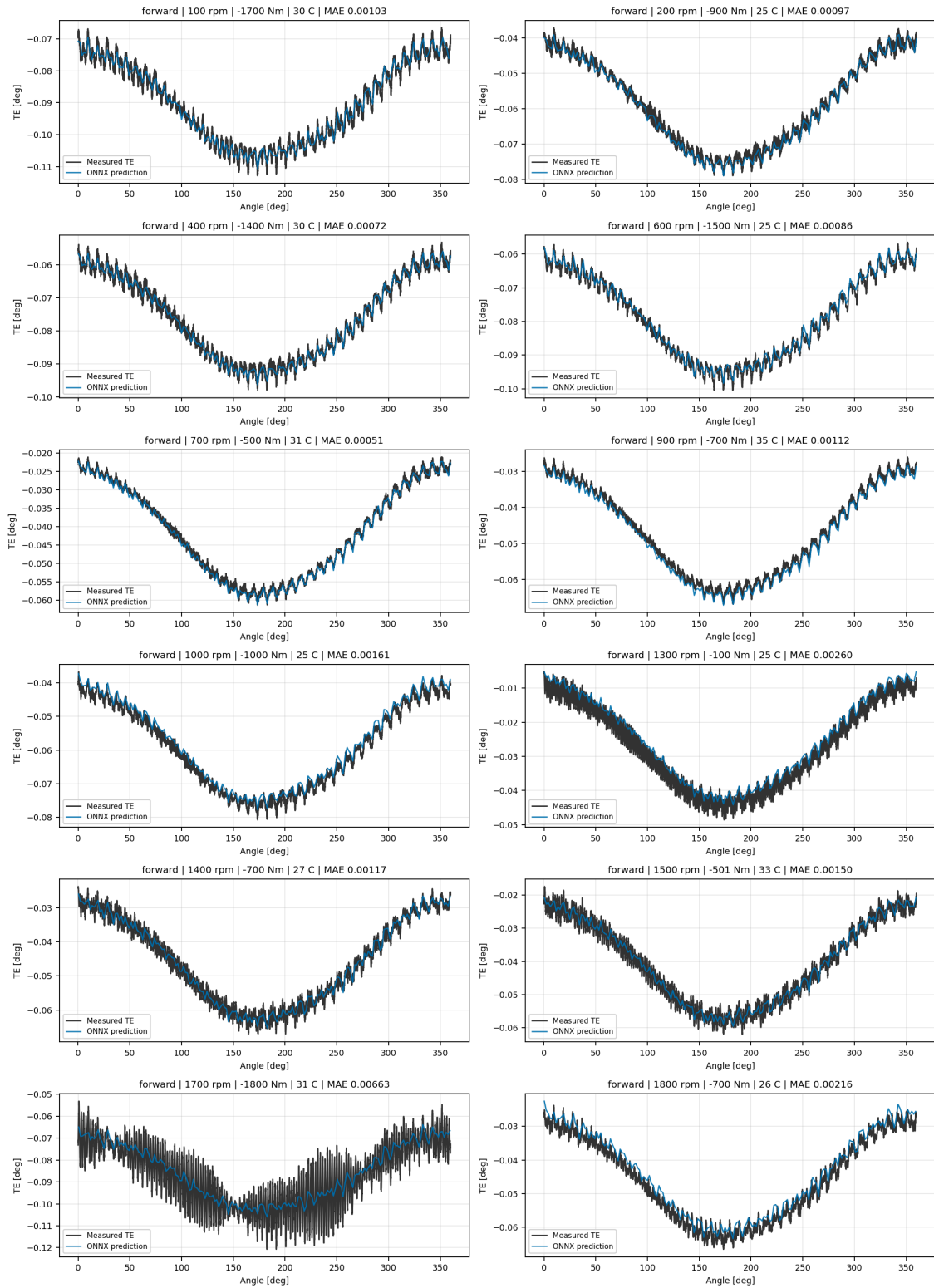
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001775	0.002136	3.859	9.256
backward	94	0.002153	0.002607	4.080	10.498
global	194	0.001958	0.002368	3.961	10.273

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000205	0.000755	0.001552	0.003671
backward	0.000237	0.000542	0.002057	0.005476
global	-0.000060	0.000645	0.001794	0.004792

Forward 12-Curve Page

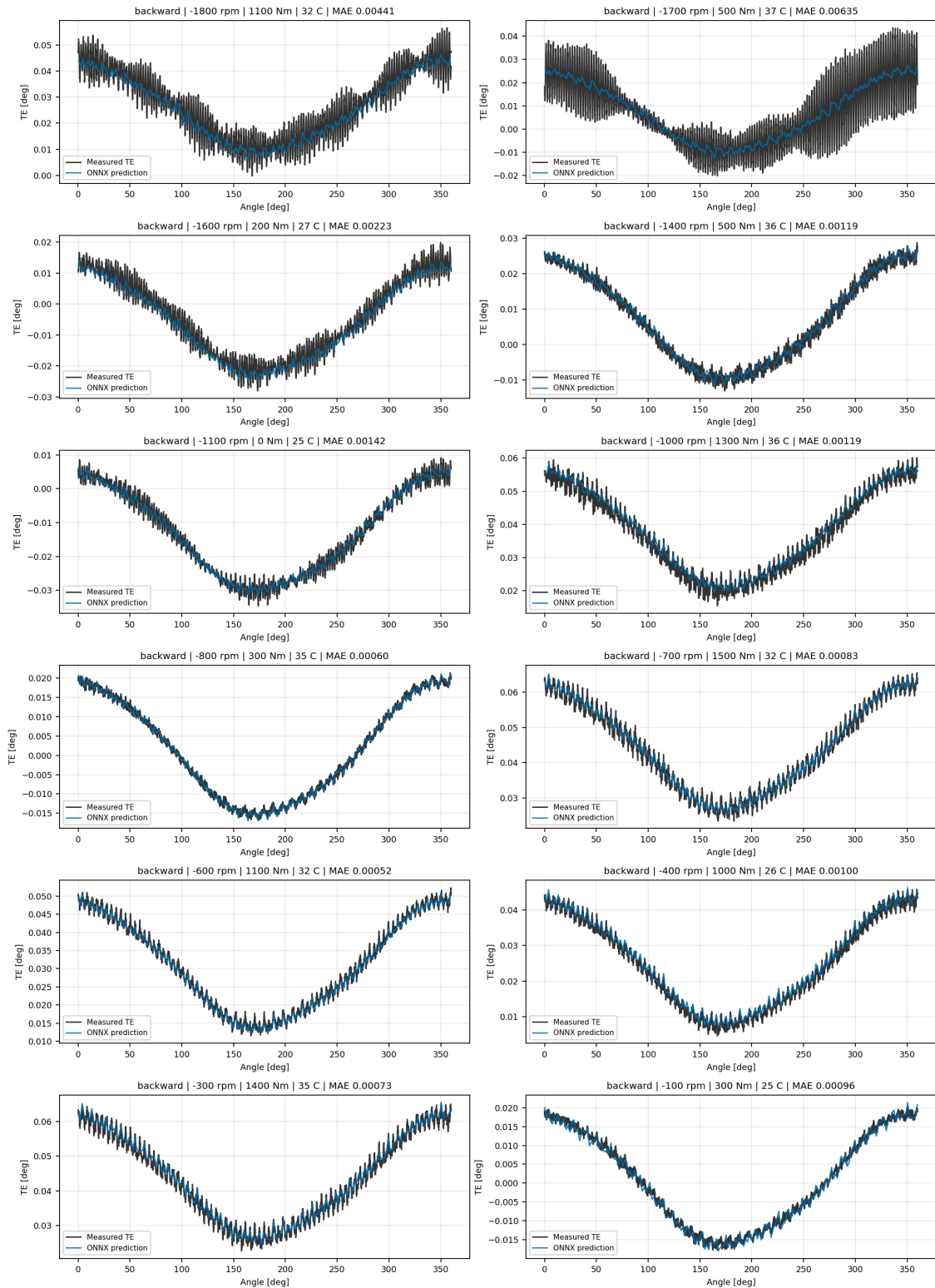
Polished Dataset + Actual Values | wave3_2_harmonic_residual_offset | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

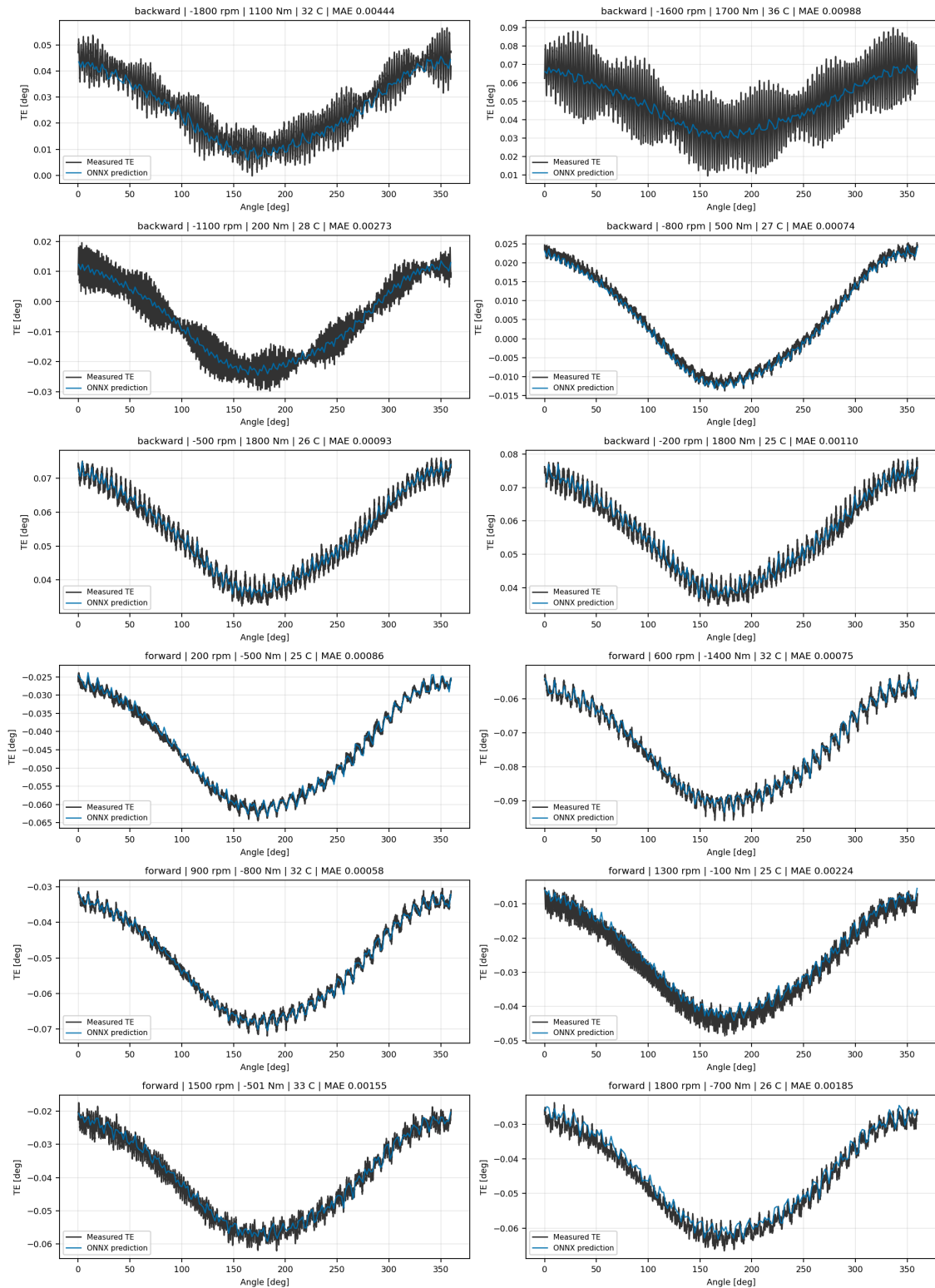
Polished Dataset + Actual Values | wave3_2_harmonic_residual_offset | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave3_2_harmonic_residual_offset | global



polished_dataset_actual_values global 12-curve collage